

Why Machine Learning Falls Short in Predictive Maintenance for Metro Trains? Real-World Challenges

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Abstract—Availability is a key performance indicator in metro rail operations but remains strongly affected by unplanned service-affecting failures (SFAs). Predictive maintenance (PdM) is widely promoted as a means to improve availability. However, its real-world effectiveness in metro systems remains unclear. This paper investigates promising supervised and unsupervised machine-learning approaches for PdM in metro vehicles, examines the impact of temporal design choices on predictive performance, and assesses their suitability for operational deployment. A systematic literature review identifies Random Forest (RF) classifiers and Long Short-Term Memory autoencoders (LSTM-AE) as representative methods for multivariate time-series data. These models are evaluated using the real-world, event-based MetroAT dataset, containing high-frequency pneumatic system sensor data. Temporal aggregation windows, sequence lengths, and early-warning label horizons are derived through data-driven signal analysis to enable a controlled and comparable evaluation. Although both approaches detect pre-failure anomalies, test-set results indicate limited deployment readiness (F_1 -score: 0.21 for RF; 0.08 for LSTM-AE), driven by high false-alarm rates and sensitivity to temporal misalignment. These results highlight fundamental challenges in applying machine-learning-based PdM to safety-critical rail systems, showing that while PdM can reveal early degradation patterns, it must overcome contextual uncertainty and operational complexity. Improving false-alarm interpretability and augmenting contextual features with event-aware information are critical next steps.

Index Terms—Availability, Rail Vehicle Maintenance, Predictive Maintenance, Machine Learning, Random Forest, LSTM Autoencoder

I. INTRODUCTION

Modern metro trains operate under increasing technical and operational complexity, with aging assets and heterogeneous usage patterns challenging conventional maintenance planning [Dersin et al., 2008]. Under these conditions, a substantial share of service-affecting failures (SFAs) occurs unexpectedly

during revenue service, leading to unplanned downtime and reduced fleet availability. Time- or rule-based maintenance strategies mainly address statistically predictable failure modes and offer limited capability to manage unforeseen disruptions [Garcia Figueiredo-Pinto et al., 2021].

Advances in onboard sensing have increased the volume and resolution of operational data in modern metro fleets, enabling data-driven maintenance approaches based on machine learning (ML). However, effective use of data remains challenging, as operational datasets are characterised by rare failure events, delayed or interval-based failure documentation, and strong temporal and contextual dependencies that are difficult to capture with conventional models [Nunes et al., 2023]. These challenges are significant in availability-critical subsystems like pneumatic air production, limiting failure signatures.

Figure 1 contextualises these challenges within an availability-oriented maintenance roadmap. Although conventional maintenance strategies have reduced reduced SFAs by approximately 67% between 2009 and 2021 [Steiner et al., 2025a], a substantial fraction of disturbances remain unresolved. Prognostic health monitoring (PHM), or predictive maintenance (PdM), seeks to address this gap through early detection of degradation and abnormal system behavior [Nunes et al., 2023]. Moreover, even when SFA-related anomalies are present in sensor data, their systematic integration into maintenance decision-making is constrained by limitations in data quality, temporal alignment, and missing event context. However, even when SFA-related anomalies are present in sensor data, their integration into maintenance decision-making is constrained by data quality, temporal misalignment, and missing event context [Zhang et al., 2022]. These limitations are most pronounced in the failure prediction phase within PdM approaches, motivating the present study.

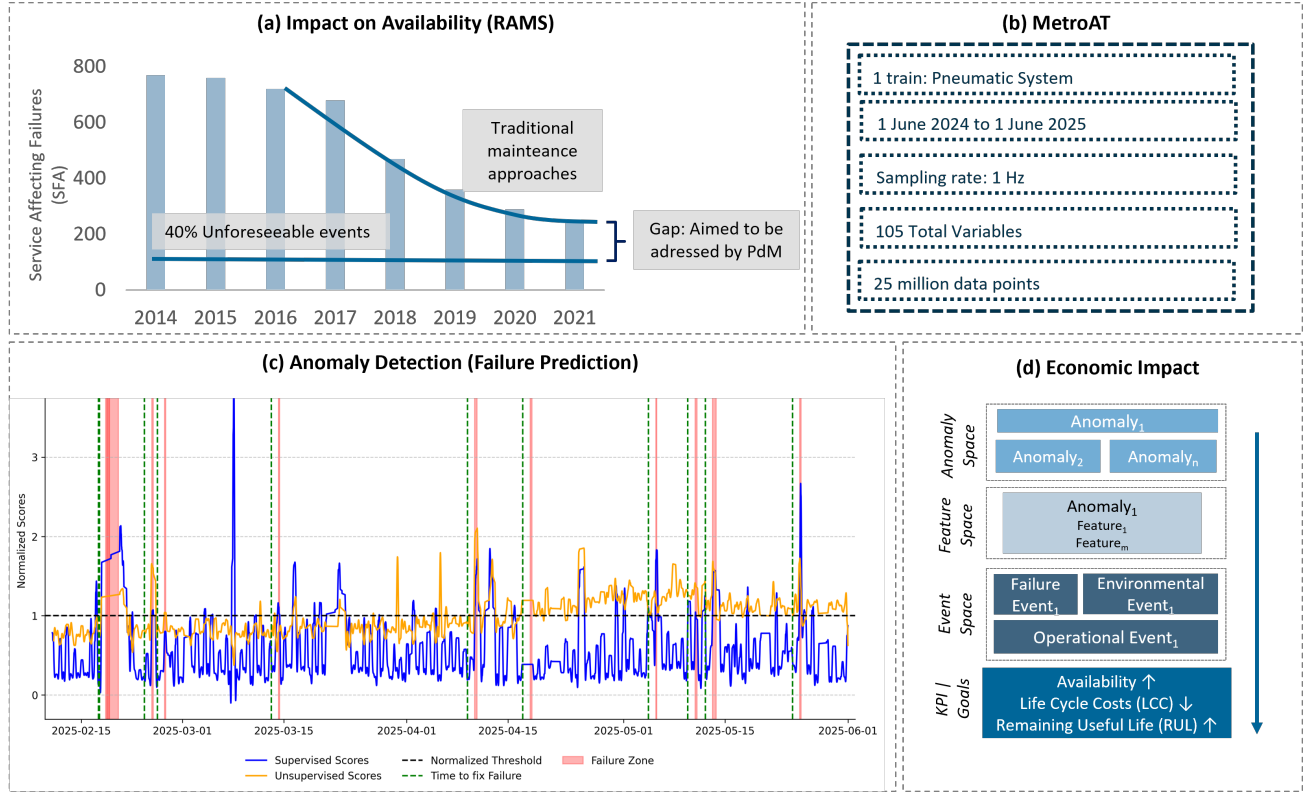


Fig. 1. PdM-based availability roadmap, illustrating (a) the remaining gap driven by service disruptions [Steiner et al., 2025a], (b) the event-based real-world MetroAT dataset [Steiner et al., 2025b], (c) full test-period results with normalised anomaly scores from the selected RF (blue) and LSTM-AE (orange), including the decision threshold (dashed) and failure intervals (red), and (d) the key steps linking PdM outputs to tangible availability impact.

II. STUDY DESIGN

Referring to the aforementioned limitations, the performance of PdM is sensitive to both the selected ML method and the chosen temporal resolution. However, this sensitivity is rarely subjected to systematic analysis, exposing key research gaps: (1) the absence of a comparative performance evaluation of how temporal resolution influences predictive performance across modeling paradigms, (2) the lack of controlled and comparable performance evaluations between supervised and unsupervised approaches under aligned temporal and labeling assumptions, and (3) the limited validation of this methods on large-scale, real-world operational datasets, which restricts conclusions about robustness and applicability. These gaps motivate the research questions of this work, summarised as follows:

- **RQ1:** Which supervised and unsupervised machine learning approaches show the greatest potential for predictive maintenance in metro vehicles?
- **RQ2:** To what extent does temporal resolution influence the technical performance, measured by the F_1 -score, of the most promising supervised and unsupervised learning approaches on the MetroAT dataset?
- **RQ3:** To what extent are these supervised and unsupervised learning approaches suitable for deployment in real-world maintenance applications?

To address **RQ1**, a Systematic Literature Review (SLR) was conducted following PRISMA 2020 guidelines [Page et al., 2021] to identify ML-based PdM approaches for multivariate time-series data in safety-critical metro systems. Searches across ACM Digital Library, IEEE Xplore, and SpringerLink were structured using the PICOC framework [Carvalho et al., 2019], resulting in seven primary studies included for synthesis. To address **RQ2**, the open-source MetroAT dataset [Steiner et al., 2025b] serves as a reference. Multiple domain-informed temporal resolutions are derived, and a controlled comparison of supervised and unsupervised approaches is conducted to assess performance stability and robustness. Finally, **RQ3** evaluates the integration of these predictions into operational settings and identifies conditions limiting their practical effectiveness.

III. SYSTEMATIC LITERATURE REVIEW AND FINDINGS

A. Service Affecting Failures and RAMS in Metro Trains

In metro operations, maintenance impact is largely driven by unplanned SAFs, which withdraw vehicles from service, disrupt scheduled maintenance, and force reactive interventions that strain workshop capacity and spare parts [Wang et al., 2025]. Even moderate SAF rates can disproportionately degrade RAMS performance, as a small number of unexpected failures dominate service delays beyond contractual limits [Appoh and Yunusa-Kaltungo, 2021]. To address these

RAMS-critical events, PdM enables early anomaly detection in SFA-relevant systems, such as pneumatic air-production and braking systems, thereby shifting maintenance from corrective to planned execution [Toribio et al., 2026]. Timely PdM alerts support depot-based interventions, reducing in-service failures (reliability), shortening restoration times (maintainability), and improving availability, while helping operators meet fleet availability and service-level requirements and mitigate SAF-related penalty costs [Kumari et al., 2025]. Prior work shows that PdM-driven decision support yields measurable RAMS benefits by reducing defect incidence and maintenance occupation times through learned condition indicators [Sresakoolchai and Kaewunruen, 2023], with subsequent studies reporting associated economic gains and improved profitability [Schenkelberg et al., 2020].

B. ML Approaches in Metro Train Maintenance

This section reviews state-of-the-art supervised and unsupervised ML approaches addressing **RQ1**, all based on the MetroPT dataset [Veloso et al., 2022].

Supervised learning: In PdM, the expected paradigm uses ground-truth failure annotations from maintenance reports. However, none of the reviewed studies employ true failure labels. Instead, proxy labels are generated via unsupervised anomaly detection methods, such as Isolation Forests or interquartile range (IQR) heuristics, and binary labels (failure/non-failure) are assigned based on detected outliers. These inferred labels train supervised classifiers. Consequently, although models follow a supervised pipeline, they learn to reproduce the anomaly detector rather than predict actual failures (see Table I).

TABLE I
COMPARING SUPERVISED LEARNING ALGORITHMS TRAINED ON METROPT FOR METRO TRAIN PREDICTIVE MAINTENANCE.

Algorithm	Characteristics
Logistic Regression [Barpute et al., 2024], [Sandhu et al., 2024]	- Input data type: Digital and analog sensor readings - Labeling strategy: Proxy labels via Isolation Forest and IQR F_1-score: 0.84–0.9980 - Temporal and imbalance handling: No temporal awareness; limited imbalance mitigation via resampling or reweighting
Random Forest [Barpute et al., 2024], [Sandhu et al., 2024]	- Input data type: Digital and analog sensor readings - Labeling strategy: Proxy labels via Isolation Forest and IQR F_1-score: 0.94–0.9960 - Temporal and imbalance handling: No temporal awareness; ensemble diversity improves minority-class learning
XGBoost [Barpute et al., 2024]	- Input data type: Digital and analog sensor readings - Labeling strategy: Proxy labels via Isolation Forest F_1-score: 0.9970 - Temporal and imbalance handling: No temporal awareness; imbalance controlled via hyperparameters
LightGBM [Barpute et al., 2024]	- Input data type: Digital and analog sensor readings - Labeling strategy: Proxy labels via Isolation Forest F_1-score: 0.9960 - Temporal and imbalance handling: No temporal awareness; strong imbalance handling through class weighting
CatBoost [Barpute et al., 2024]	- Input data type: Digital and analog sensor readings - Labeling strategy: Proxy labels via Isolation Forest F_1-score: 0.9960 - Temporal and imbalance handling: No temporal awareness; internal mechanisms for class imbalance

^aThe selected approach is highlighted in **bold**.

Unsupervised Learning approaches are well suited to PdM in metro systems due to scarce labeled failure data. Rather than relying on ground-truth annotations, they identify patterns and deviations from normal operation [Chandola et al., 2009]. In practice, many reviewed studies implement what is more accurately described as semi-supervised anomaly detection, where models are trained on normal data and later used to flag deviations as potential faults. Although true semi-supervised learning assumes limited labeled examples, the term is frequently used interchangeably with unsupervised learning in the PdM literature [Chapelle et al., 2006] (see Table II).

TABLE II
COMPARISON OF UNSUPERVISED LEARNING ALGORITHMS TRAINED ON THE METROPT DATASET FOR METRO TRAIN PREDICTIVE MAINTENANCE.

Algorithm	Characteristics
Stacked Autoencoder [Davari et al., 2021]	- Input data type: Digital, analog; combined sensor readings - Labeling strategy: Trained on normal data; evaluated on known failures F_1-score: 0.45–0.7794 - Temporal and imbalance handling: Cycle-based segmentation; imbalance-robust via normal-only training
Variational Autoencoder (VA) [Davari et al., 2021]	- Input data type: Digital or analog sensor readings - Labeling strategy: Trained on normal data; evaluated on known failures F_1-score: 0.39–0.59 - Temporal and imbalance handling: No explicit temporal modeling; robust imbalance handling via generative prior
Long Short-Term Memory Autoencoder (LSTM-AE) [Davari et al., 2024]	- Input data type: Digital and analog sensor readings - Labeling strategy: Trained on normal data; evaluated on known failures F_1-score: 0.9743 - Temporal and imbalance handling: Strong temporal modeling via LSTM; robust to imbalance through normal-only learning
Convolutional Neural Network Autoencoder (CNN-AE) [Zafra et al., 2025]	- Input data type: Forecasted sensor readings from CNN-LSTM - Labeling strategy: Trained on normal data; applied to forecasted inputs F_1-score: 0.66–0.7539 - Temporal and imbalance handling: Medium temporal awareness via convolutions; robust imbalance handling through normal-only training

^aThe selected approach is highlighted in **bold**.

As summarised in Table I, supervised failure prediction relies primarily on aggregated features and lacks explicit temporal modeling. In contrast, unsupervised approaches (Table II) incorporate temporal structure through heterogeneous windowing strategies. Based on the comparative analysis of the datasets and feature usage, the following findings are derived: (1) The MetroPT dataset [Veloso et al., 2022] is relatively limited in complexity compared to real-world operational settings, where sensor data typically exhibit higher noise levels, missing records, and irregular logging. Consequently, conclusions drawn from MetroPT may not fully generalise, motivating the evaluation of PdM methods on more complex datasets (*e.g.*, MetroAT [Steiner et al., 2025b]), (2) Supervised learning approaches often lack detailed characterisations of temporal characteristics, and window-based feature extraction is largely absent.

Selecting benchmark ML approaches was non-trivial, as su-

pervised methods exhibit similar reported performance, which is itself questionable due to reliance on proxy labels. Nevertheless, Random Forest (RF) was selected as the supervised baseline because its ensemble structure naturally mitigates skewed class distributions, making it suitable for real-world PdM scenarios where failures are rare. Among unsupervised methods, LSTM-based Autoencoders (LSTM-AE) were included to explicitly capture temporal dependencies.

IV. PROPOSED APPROACH

This section describes the proposed approach by outlining the steps involved in deriving ML outcomes for PdM: (1) the selection of temporal windows, (2) feature generation, and (3) the definition of prediction targets [Helm, 2026].

A. Data-driven temporal windows

A well chosen temporal window ensures that meaningful signal dynamics are included while avoiding excessive noise or irrelevant variability, thereby improving anomaly detection and prediction accuracy [Barpute et al., 2024], [Davari et al., 2024]. Therefore, this work includes three complementary temporal design choices:

- 1) The **feature aggregation window** for the RF determines how high-frequency sensor data are condensed into dynamic features (*e.g.* mean, variance, gradients) that capture fluctuations and retain localised precursors.
- 2) The **sequence window** for the LSTM-AE defines the temporal context (*i.e.*, number of consecutive time steps) over which the model learns stable normal-operation dynamics and detects deviations [Davari et al., 2021].
- 3) The **failure label shift window** defines the prognostic horizon of the RF, aligning pre-failure patterns with subsequent failure events to support early failure prediction.

To derive feature aggregation and sequence window lengths, the analysis focuses on the main reservoir pressure signal (*MAIN_RESERVOIR_PRESSURE*) recorded for both control wagons (*i.e.*, CW1 and CW2). As the most direct indicator of the pneumatic system, it exhibits fast fluctuations (compressor cycles) and slower operational rhythms (station-to-station travel) [Zafra et al., 2025]. Before estimating time scales, the signal is stationarised by per-day demanding to remove slow baseline shifts while preserving the full-day coverage:

$$X_t^{(\text{dm})} = X_t - \mathbb{E}[X_t | \text{day}(t)], \quad (1)$$

where X_t reflects the original signal at time t and $\mathbb{E}[X_t | \text{day}(t)]$ is the mean of the signal on the corresponding day. The characteristic time scales are then extracted from $X_t^{(\text{dm})}$.

Autocorrelation Decay (decorrelation time): For a time series X_t , the *Autocorrelation Function (ACF)* at lag τ is

$$\rho(\tau) = \frac{\mathbb{E}[(X_t - \mu)(X_{t+\tau} - \mu)]}{\text{Var}(X_t)}. \quad (2)$$

For the demeaned signal $X_t^{(\text{dm})}$, we have $\mu = 0$, and the ACF is therefore computed directly from $X_t^{(\text{dm})}$. The sample

ACF, denoted by r_k , is estimated using an FFT-based estimator and normalised such that $\rho(0) = 1$. Following Bartlett's approximation, the standard error of the sample ACF at lag k is given by:

$$\text{SE}_k \approx \sqrt{\frac{1 + 2 \sum_{j=1}^{k-1} r_j^2}{N}}, \quad (3)$$

where N is the number of samples. The decorrelation time τ_{corr} is defined as the first lag k at which the sample ACF enters the 95% *Bartlett-corrected confidence band*,

$$|r_k| \leq 1.96 \text{SE}_k \Rightarrow \tau_{\text{corr}} = k \Delta t, \quad (4)$$

where Δt is the sampling interval [Parzen, 1963]. Intuitively, τ_{corr} represents the time horizon beyond which the signal becomes effectively uncorrelated with its past, providing a principled, data-driven estimate of the system's temporal memory. For the full dataset, the *Bartlett-corrected 95% confidence band* crossing occurs at $\tau_{\text{corr}} \approx 701\text{s}$ (about 11.5 minutes). When calculating decorrelation times on a per-day basis, we obtain a median $\tilde{\tau}_{\text{corr}} \approx 410\text{s}$, with an IQR of [343s, 499s]. The narrow confidence bands result from the very high sample count ($N > 20$ million), as illustrated in Figure 2.

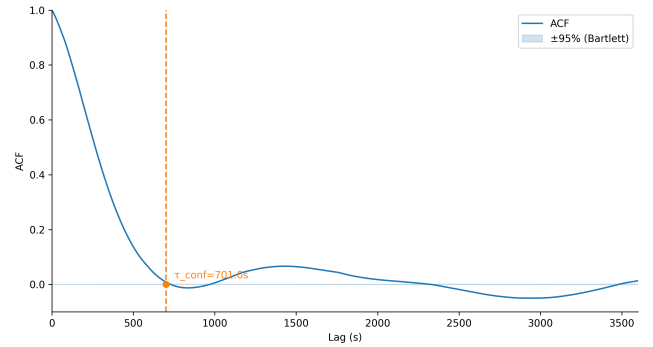


Fig. 2. Autocorrelation of stationarised (*MAIN_RESERVOIR_PRESSURE*) (full dataset) with *Bartlett-corrected 95% confidence band*.

Failure Precursors: For each documented failure, the (*MAIN_RESERVOIR_PRESSURE*) is analysed in a lookback window of 24 hours preceding the official failure timestamp t_f . A rolling variance is computed using a sliding window of length equal to the decorrelation time τ_{corr} ,

$$v_t = \frac{1}{\tau_{\text{corr}}} \sum_{i=t-\tau_{\text{corr}}+1}^t (X_i - \bar{X}_{t, \tau_{\text{corr}}})^2, \quad (5)$$

where $\bar{X}_{t, \tau_{\text{corr}}}$ is the mean of X over the window. The variance trace v_t is standardised into a z-score, and anomalies are defined above a threshold of $z \geq 2$. The precursor onset t_p is taken as the earliest such exceedance within the lookback horizon that occurs at least continuously for $0.5 \tau_{\text{corr}}$ before t_f . The precursor horizon τ_{lead} quantifies the lead time at which abnormal dynamics typically become observable relative to the official failure time,

$$\tau_{\text{lead}} = t_f - t_p, \quad (6)$$

The extracted time scales (τ_{corr} and τ_{lead}) are mapped to candidate window lengths for the respective modeling tasks. The autocorrelation-derived time scale τ_{corr} is employed to define both the feature aggregation window and the LSTM-AE sequence window, thereby ensuring consistency between supervised feature engineering and unsupervised sequence modeling. The resulting raw window lengths are subsequently rounded to canonical durations to improve interpretability and alignment with operational cycles:

$$\begin{aligned} W_{\text{feat}} &\in \left\{ \frac{1}{2} \tau_{\text{corr}}, \tau_{\text{corr}}, 2 \tau_{\text{corr}} \right\}, \\ W_{\text{seq}} &\in \left\{ \frac{1}{2} \tau_{\text{corr}}, \tau_{\text{corr}}, 2 \tau_{\text{corr}} \right\}, \\ W_{\text{feat}}, W_{\text{seq}} &\in \{6 \text{ min}, 11.5 \text{ min}, 23 \text{ min}\} \end{aligned} \quad (7)$$

Failure Label Shift Window: Across the 11 documented failures, the detection pipeline identified consistent precursor dynamics in the rolling-variance analysis. The median precursor lead time was estimated as $\tau_{\text{lead}} \approx 6.2\text{h}$, with an interquartile range (IQR) of $[2.9\text{h}, 15.5\text{h}]$. As shown in Figure 3, precursor signals systematically emerge several hours prior to the official failure annotations. Motivated by the empirical distribution of τ_{lead} , a discrete set of candidate failure label shift windows were defined:

$$W_{\text{label}} \in \left\{ 0 \tau_{\text{lead}}, \frac{1}{2} \tau_{\text{lead}}, \tau_{\text{lead}} \right\} = \{0\text{h}, 3.1\text{h}, 6.2\text{h}\} \quad (8)$$

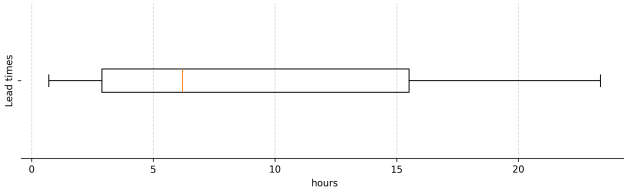


Fig. 3. Distribution of lead times τ_{lead} detected across failures

B. Feature Aggregation

For each window $W \in W_{\text{feat}} = \{6 \text{ min}, 11.5 \text{ min}, 23 \text{ min}\}$, a set of time-local features is extracted to summarise recent operating conditions. For every analog channel (e.g., pressures, brake force, speed, temperature), four time-domain aggregates are computed within the window: min, max, mean, and sum. For binary or digital channels (e.g., compressor running, brake command, pressure-available flags), two indicators are derived: (i) *active_s*, the total number of seconds the signal equals 1 within the window (duty cycle), and (ii) *flips*, the number of state transitions ($0 \leftrightarrow 1$), capturing switching intensity.

Overall, the extracted features are predominantly statistical time-domain summaries, instead of explicitly computing frequency-domain descriptors, temporal structure is captured through the multi-scale design of W_{feat} , which represents short- and medium-term dynamics via parallel windows of different lengths. This feature design is consistent with established predictive-maintenance practice, where simple time-domain descriptors often provide strong baselines [Barapatre et al., 2024].

C. Prediction Targets

The prediction objective is to determine whether a failure will occur within a data-driven early-warning horizon. Therefore, three feature horizons W_{feat} are considered, together with a non-zero label horizon.

In the supervised setting, separate RF classifiers are trained and evaluated for each early-warning horizon. To prevent ambiguous training samples caused by label shifting, windows are retained only when the shifted label indicates an upcoming failure or when both the current and shifted states indicate normal operation. Windows that fall within an ongoing failure without a failure occurring within the shifted horizon are excluded. This filtering strategy ensures a well-defined early-warning target and avoids trivial negative samples.

In the unsupervised setting, the LSTM-AE is trained using windows corresponding to normal operation, and during inference anomaly scores are computed from reconstruction errors on unseen data. For comparability with the supervised approach, alarms generated on the temporal hold-out set are evaluated against SFAs using the same early-warning horizons. An alarm is considered correct if it occurs within the corresponding pre-failure window, enabling a direct comparison of precision (P), recall (R), and F_1 -score between the RF and LSTM-AE on the same test period.

V. RESULTS

To ensure a fair out-of-sample comparison between the supervised and unsupervised approaches, a strictly temporal data split of *MetroAT* is employed. The initial 70% of the timeline is used as a development set for model training, hyperparameter tuning, and threshold selection, and contains only information available prior to evaluation. The remaining 30% of the timeline, spanning approximately 110 days from mid-February 2025 to June 2025, is reserved as an untouched hold-out test set.

A. Results at Varying Temporal Resolutions

The aggregation or sequence window (W_{feat} for RF and W_{seq} for LSTM-AE) as well as the labeling window W_{label} are systematically varied, and performance is evaluated using F_1 , P, and R. To reduce noise and therefore false alarms in minute-level anomaly scores, a *zero-phase fourth-order digital Butterworth low-pass filter* (LPF) is applied to the prediction or reconstruction error sequence. The LPF is parameterised by a sampling rate of $f_s = 1 \text{ Hz}$ and a cutoff frequency f_c , which is swept over the range $[0.01, 0.1]$ for each selected model configuration, following Davari et al. [Davari et al., 2021]. In parallel, the decision threshold θ applied to the filtered anomaly scores is varied over the range $[0.01, 0.99]$. The optimal pair (f_c, θ) is selected by maximising the F_1 -score on the development set and used for evaluation on the hold-out test set.

Supervised Model (RF): Table III summarises the temporal and LPF hyperparameters of the RF variants together with their performance metrics, empirically identifying the best-performing hyperparameter settings. It shows that the F_1

score ranges from 0.62 to 0.98 across temporal configurations. The strongest setting achieves $F_1 = 0.98$ with high P and perfect R ($P = 0.96$, $R = 1.0$) under light LPF smoothing ($f_c = 0.0128$, $\theta = 0.15$), whereas the weakest configuration attains only $F_1 = 0.62$ despite perfect R ($R = 1.0$), primarily due to a substantial drop in precision ($P = 0.45$). High-performing models require only mild LPF smoothing ($f_c \approx 0.01$ – 0.02), whereas weaker configurations depend on substantially stronger smoothing and higher thresholds yet still yield inferior F_1 scores, indicating that temporal alignment of W_{feat} with the underlying process dynamics is the primary driver of performance rather than LPF tuning.

TABLE III
RF DEVELOPMENT SET RESULTS ACROSS TEMPORAL WINDOWS.

W_{feat}	W_{label}	Decision Basis (Development Set)				
		F_1	P	R	LPF f_c^*	θ^*
6 min	6.2 h	0.98	0.96	1.00	0.012755	0.15
11 min	3.1 h	0.96	0.92	1.00	0.012755	0.16
23 min	3.1 h	0.96	0.92	1.00	0.019184	0.17
11 min	6.2 h	0.92	0.92	0.92	0.010918	0.16
6 min	3.1 h	0.75	0.96	0.61	0.010918	0.17
23 min	0 h	0.71	0.71	0.71	0.047653	0.45
11 min	0 h	0.66	0.57	0.80	0.027449	0.15
23 min	6.2 h	0.66	0.50	1.00	0.039388	0.18
6 min	0 h	0.62	0.45	1.00	0.013673	0.12

^aBest result highlighted in **bold**.

Unsupervised Model (LSTM-AE): Table IV summarises the temporal and LPF hyperparameters of the selected LSTM-AE configurations, with F_1 scores ranging from 0.20 to 0.75. The best-performing setting uses a 23 min sequence and achieves $F_1 = 0.75$ with balanced $P = 0.64$ and $R = 0.90$ under moderate smoothing ($f_c = 0.0486$, $\theta = 0.60$), indicating that alignment of the sequence length W_{seq} with the system’s intrinsic temporal scale (approaching $2\tau_{\text{corr}}$) is the primary determinant of performance. In contrast, short-sequence configurations (~ 6 min) consistently underperform across label horizons, yielding F_1 scores as low as 0.20 despite comparable or weaker smoothing, as insufficient temporal context amplifies reconstruction-error jitter and fragments early failure precursors.

TABLE IV
LSTM-AE DEVELOPMENT SET RESULTS ACROSS TEMPORAL WINDOWS.

W_{seq}	W_{label}	Decision Basis (Development Set)				
		F_1	P	R	LPF f_c^*	θ^*
23 min	3.1 h	0.75	0.64	0.90	0.0486	0.60
23 min	6.2 h	0.70	0.70	0.70	0.0679	0.65
11 min	6.2 h	0.63	0.85	0.50	0.0302	0.63
11 min	3.1 h	0.58	0.90	0.43	0.0357	0.72
6 min	6.2 h	0.47	0.51	0.43	0.0201	0.70
6 min	3.1 h	0.41	0.43	0.40	0.0183	0.65
23 min	0 h	0.31	0.33	0.29	0.0660	0.90
11 min	0 h	0.28	0.29	0.27	0.0274	0.58
6 min	0 h	0.20	0.16	0.25	0.0100	0.43

^aBest result highlighted in **bold**.

B. Test Set Cross-Comparison

Two caveats are relevant when comparing the ML approaches: (i) sequence models reduce effective evaluation coverage due to windowing, so absolute counts (True Negative (TN)/ False Positive (FP)) are not directly comparable; (ii) operational costs are asymmetric, as false alarms (= FP) are typically expensive. Under these conditions, P-oriented models are generally preferable for deployment, while R-oriented models may serve as sentinels when the priority is to capture nearly all failures. On the hold-out test set, the RF achieves a substantially higher F_1 score (0.21 vs. 0.08) by combining higher P (0.14 vs. 0.04) with moderate R (0.37), whereas the LSTM-AE attains very high R (0.92) at the expense of many FP, which markedly depresses F_1 (Table V).

TABLE V
TEST-SET PERFORMANCE COMPARISON.

Performance Metric	RF	LSTM-AE
F_1	0.21	0.08
P	0.14	0.04
R	0.37	0.92
TN (% of negatives)	92.25%	56.58%
FP (% of negatives)	7.75%	43.42%
FN (% of positives)	62.71%	8.05%
TP (% of positives)	37.29%	91.95%

^aBetter value in **bold**. TN/FP are % of negatives; FN/TP are % of positives.

^bClass totals - RF: #Neg=18513, #Pos=641

^cClass totals - LSTM-AE: #Neg=4330, #Pos=87

While these aggregate metrics are standard for classification, they do not capture when alarms occur or how long they persist, factors that are critical in operational settings. To assess practical utility, the following analysis compares score trajectories over time (RF posterior vs. LSTM-AE reconstruction error) alongside shaded failure intervals, and quantifies lead time and early-hit rates (*e.g.*, detections within $[-6\text{ d}, -1\text{ d}]$).

C. Time-Resolved Comparison of Scores

Over the full evaluation period (Figure 1(c)), the LSTM-AE issues early alarms for 9 of 11 failures with 14 false alarms, whereas RF achieves perfect early-hit coverage (11/11) within the $[-6\text{ d}, -1\text{ d}]$ window with the same number of false alarms. A closer time-series analysis reveals three characteristic patterns: (1) RF triggers earlier by exploiting asset-specific precursors learned from historical labels, (2) LSTM-AE captures gradual distributional drift before changes become discriminative in the supervised feature space, and (3) divergent responses reflect complementary sensitivity to distinct failure precursors over time. Notably, the LSTM-AE shows a sustained elevation after 2025-04-17, consistent with a post-shift reconstruction-error baseline. This behavior motivates periodic retraining and/or adaptive thresholding to mitigate alarm persistence.

VI. DISCUSSION

A. Research Findings

RQ1: Table I and II show that metro focused PdM research predominantly relies on supervised classifiers and unsuper-

vised reconstruction. Public corpora (e.g. *MetroPT*) remain widely used, while operational datasets that include event intervals and asset context are scarce. Temporal design aspects such as windowing or label horizons are often underspecified, and asset history features are underrepresented.

RQ2: A core finding is that time-related design choices play a central role in the performance of both supervised and unsupervised models. As shown in Tables III and IV, these two approaches favor distinct temporal configurations: RF peaks with fine-grained feature windows ($W_{\text{feat}} = 6\text{min}$) and longer anticipatory label shifts ($W_{\text{label}} \approx 6.2\text{h}$), whereas LSTM-AE performs best with longer sequences ($W_{\text{seq}} = 23\text{min}$) and shorter horizons ($W_{\text{label}} \approx 3.1\text{h}$). This divergence reflects their different use of temporal information, with RF exploiting localised discriminative fluctuations and LSTM-AE requiring extended context to learn stable representations of normal system dynamics. Furthermore, label shifting consistently improves F_1 compared to unshifted baselines, indicating the presence of detectable precursors prior to annotated failure onsets. This confirms strong temporal sensitivity and supports the design of lead-time-aware labeling strategies.

RQ3: The results indicate that both supervised and unsupervised approaches are conditionally suitable for deployment in real-world maintenance settings, but in complementary rather than standalone roles. Under realistic, event-aware evaluation, absolute performance is reduced (RF: $F_1 = 0.21$; LSTM-AE: $F_1 = 0.08$), reflecting the challenges of operational data, yet yielding deployment-relevant insights (see Table V). While the RF achieves higher F_1 through greater P at comparable R , the LSTM-AE triggers earlier in several cases and highlights distinct precursor patterns. Accordingly, the RF acts as a P-oriented trigger when actionable reliability is crucial (e.g. allocating inspection resources), while the LSTM-AE functions as a broad-scope sentinel layer that raises earlier, exploratory alerts. This contrast is best interpreted as a P-R trade-off with complementary diagnostic value. Flagged intervals from either model can be used to zoom into test data, inspect contributing sensors, and identify recurring patterns behind FPs and true early hits, forming a practical pipeline for domain-driven feature design.

B. Limitations

First, the analysis of temporal windows and prediction horizons was conducted using a single, representative pneumatic pressure channel (*MAIN_RESERVOIR_PRESSURE*). This choice was guided by domain expertise identifying the signal as one of the most failure-relevant sensors. The results confirm that it is indeed predictive when paired with an appropriate temporal design. However, this also implies that conclusions regarding optimal window and horizon lengths are, strictly speaking, sensor-specific.

Secondly, comparisons with *MetroPT* results should be treated cautiously, as many RF baselines rely on proxy labels and omit asset-context variables. The lower F_1 -score observed here is therefore consistent with the greater variability and documentation gaps of an operational dataset rather than model

shortcomings. Instead, it highlights the increased difficulty of real-world deployment and the need for stronger temporal calibration and richer context modeling. Performance remains sensitive to the joint calibration of W_{feat} , W_{seq} , W_{label} , the smoothing cutoff f_c , and the threshold θ . While systematic sweeps were performed, alternative operating points may shift the P-R trade-off.

Lastly, the effectiveness and required sophistication of PdM strategies increase with system and data complexity, introducing fundamental limitations for real-world applications. While simple or weakly coupled subsystems (e.g., door mechanisms) can often be monitored using limited sensing and low-complexity models. As complexity grows, interactions between subsystems become nonlinear, operating conditions overlap, and failure modes propagate across components and noise increases [Lee and Pan, 2017]. In addition, in such complex systems, the boundaries of normal operational conditions are often ill-defined and context-dependent, leading to increased noise, uncertainty and ambiguity in fault interpretation [Nunes et al., 2023].

C. Economic Impact and Future Work

PdM does not remove part replacement or labor costs, but shifts work from unplanned failures to planned maintenance windows, reducing downtime, secondary damage, and SFAs. Its value depends on controlling false alarms (as seen in Figure 1(d)), since too many alerts increase inspections and costs, so thresholds should reflect cost trade-offs rather than accuracy alone. Moreover, benefits are realised only when PdM is integrated into maintenance planning and execution through traceable digital workflows [Florian et al., 2021]. Looking ahead, the following approaches are recommend to extend existing ML-based approaches for real-world applications: (1) identified anomalies should be enriched with event-aware information, including maintenance records, environmental and load conditions, and fleet topology, while carefully evaluating whether such asset-related context improves predictive performance without introducing data leakage, (2) hybrid-based approaches, as the combination of RF and LSTM-AE approaches could be further explored through different fusion strategies, such as logical rules, stacking, or model averaging. The RF might serve as the primary trigger for maintenance actions, while LSTM-AE is monitored as a high-R indicator to flag slow drifts and to prioritise investigations during the early-warning window, (3) analysing feature values in TP and FP sequences using feature attribution methods could uncover richer contextual insights, which may in turn inform the design of more interpretable and robust features (i.e., domain-aware feature labels) for future models and (4) finally, this work focuses on binary early-warning detection and does not address Remaining Useful Life (RUL) estimation, failure-type classification, or per-instance explainability. Extending the framework to include RUL regression and multi-label failure-type classification could enable a more detailed understanding of degradation behavior and support more targeted maintenance decisions beyond simple early-warning signals.

VII. CONCLUSION

This work demonstrates that machine-learning-based PdM is both feasible and operationally valuable when temporal design choices and asset context are treated as first-class modeling objects. The expected objectives highlighted: (i) a structured and comprehensive overview of state-of-the-art PdM approaches for metro vehicles, (ii) a systematic head-to-head comparison of supervised and unsupervised ML models on the *MetroAT* dataset, with particular emphasis on the correct specification of temporal and LPF hyperparameters, and (3) highlighting the inherent complexity of applying ML to real-world operational data.

While some existing studies report strong performance metrics, these results often rely on implicit dataset preselection or simplifying assumptions that may not hold under real deployment conditions. Rare failure events, heterogeneous assets, and evolving operational conditions limit the effectiveness of purely data-driven PdM solutions on real-world applications. Without additional domain knowledge or structural constraints, a only ML-driven approach appears unrealistic, as critical events occur too infrequently to be reliably learned from data alone.

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REFERENCES

- [Appoh and Yunusa-Kaltungo, 2021] Appoh, F. and Yunusa-Kaltungo, A. (2021). Composite hybrid framework for through-life multi-objective failure analysis and optimisation. *IEEE Access*, 9:71505–71520.
- [Barapatre et al., 2024] Barapatre, S., Gangurde, O., Bibwe, R., and Tiwaskar, S. (2024). Predictive maintenance framework for urban metro vehicles. In *2024 5th International Conference for Emerging Technology (INCET)*, pages 1–5.
- [Barpute et al., 2024] Barpute, J. V., Suryawanshi, S., Kshirsagar, V., Bhosale, D., Patil, P., and Patil, A. (2024). Predictive Maintenance of a Metro’s Air Compressor. In *5th International Conference on Electronics and Sustainable Communication Systems (ICESC)*, pages 247–252, Coimbatore, India. IEEE.
- [Carvalho et al., 2019] Carvalho, T. P., Soares, F. A. A. M. N., Vita, R., da P. Francisco, R., Basto, J. P., and Alcalá, S. G. S. (2019). A systematic literature review of machine learning methods applied to predictive maintenance. *Computers & Industrial Engineering*, 137:106024.
- [Chandola et al., 2009] Chandola, V., Banerjee, A., and Kumar, V. (2009). Anomaly detection: A survey. *ACM Comput. Surv.*, 41(3).
- [Chapelle et al., 2006] Chapelle, O., Schölkopf, B., and Zien, A., editors (2006). *Semi-Supervised Learning*. Adaptive Computation and Machine Learning. MIT Press, Cambridge, USA.
- [Davari et al., 2024] Davari, N., Veloso, B., Ribeiro, R. P., and Gama, J. (2024). Detecting and Explaining Anomalies in the Air Production Unit of a Train. In *Proceedings of the 39th ACM/SIGAPP Symposium on Applied Computing (SAC)*, pages 358–364, Avila, Spain. ACM.
- [Davari et al., 2021] Davari, N., Veloso, B., Ribeiro, R. P., Pereira, P. M., and Gama, J. (2021). Predictive Maintenance based on Anomaly Detection using Deep Learning for Air Production Unit in the Railway Industry. In *Proceedings of the IEEE 8th International Conference on Data Science and Advanced Analytics (DSAA)*, pages 1–10, Porto, Portugal. IEEE.
- [Dersin et al., 2008] Dersin, P., Peronne, A., and Arroum, C. (2008). Selecting test and maintenance strategies to achieve availability target with lowest life-cycle cost. In *2008 Annual Reliability and Maintainability Symposium*, pages 301–306.
- [Florian et al., 2021] Florian, E., Sgarbossa, F., and Zennaro, I. (2021). Machine learning-based predictive maintenance: A cost-oriented model for implementation. *International Journal of Production Economics*, 236:108114.
- [Garcia Figueiredo-Pinto et al., 2021] Garcia Figueiredo-Pinto, D., Fan, I.-S., and Teixeira Mendes Abrahão, F. (2021). Operational availability optimization model based on the integration of predictive and scheduled maintenance. In *Proceedings of the PHM Society European Conference*, volume 6. PHM Society.
- [Helm, 2026] Helm, S. (2026). Pattern recognition for predictive maintenance in metro trains: A comparative analysis of supervised and unsupervised machine learning approaches on time-series data. Unpublished diploma thesis, Data Science.
- [Kumari et al., 2025] Kumari, J., Karim, R., and Khanna, P. (2025). Dynamic maintenance policy development for railway rolling stock. *Journal of Quality in Maintenance Engineering*, 31(5):23–42.
- [Lee and Pan, 2017] Lee, D. and Pan, R. (2017). Predictive maintenance of complex system with multi-level reliability structure. *International Journal of Production Research*, 55(16):4785–4801.
- [Nunes et al., 2023] Nunes, P., Santos, J., and Rocha, E. (2023). Challenges in predictive maintenance – a review. *CIRP Journal of Manufacturing Science and Technology*, 40:53–67.
- [Page et al., 2021] Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., McGuinness, L. A., Stewart, L. A., Thomas, J., Tricco, A. C., Welch, V. A., Whiting, P., and Moher, D. (2021). The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews. *BMJ*, page 9.
- [Parzen, 1963] Parzen, E. (1963). On Spectral Analysis with Missing Observations and Amplitude Modulation. *Sankhyā: The Indian Journal of Statistics*, 25(4):383–392.
- [Sandhu et al., 2024] Sandhu, J., Mahapatra, B., Kulkarni, S., and Bhatt, A. (2024). MetroPT Predictive Maintenance Using Logistic Regression and Random Forest with Isolation Forest Preprocessing. In *Proceedings of the 12th International Conference on Soft Computing for Problem Solving*, pages 503–513, Singapore. Springer.
- [Schenkelberg et al., 2020] Schenkelberg, K., Seidenberg, U., and Ansari, F. (2020). Supervised machine learning for knowledge-based analysis of maintenance impact on profitability. *IFAC-PapersOnLine*, 53(2):10651–10657.
- [Sresakoolchai and Kaewunruen, 2023] Sresakoolchai, J. and Kaewunruen, S. (2023). Railway infrastructure maintenance efficiency improvement using deep reinforcement learning integrated with digital twin based on track geometry and component defects. *Scientific Reports*, 13(1):2439.
- [Steiner et al., 2025a] Steiner, A., Abdelkader, O., Ansari, F., and Kollegger, A. (2025a). Datengetriebene instandhaltung von schienenfahrzeugen im ÖPNV. In *Innovative Instandhaltung – Wertschöpfungsorientierte Transformation mit und durch KI*, pages 95–106. TÜV-Verlag.
- [Steiner et al., 2025b] Steiner, A., Abdelkader, O., Helm, S., and Ansari, F. (2025b). METROAT: A benchmark dataset for data-driven maintenance in metro operations. Unpublished manuscript.
- [Toribio et al., 2026] Toribio, L., Veloso, B., Gama, J., and Zafra, A. (2026). A two-stage framework for early failure detection in predictive maintenance: A case study on metro trains. *Neurocomputing*, 670:132506.
- [Veloso et al., 2022] Veloso, B., Ribeiro, R. P., Gama, J., and Pereira, P. M. (2022). The MetroPT dataset for predictive maintenance. *Scientific Data*, 9(1):764.
- [Wang et al., 2025] Wang, Y., Li, Y., Chen, X., and Zhou, Y. (2025). Urban rail transit disruption management: a decade of research on network resilience, internal, and external strategies. *Intelligent Transportation Infrastructure*, 4:liaf014.
- [Zafra et al., 2025] Zafra, A., Veloso, B., and Gama, J. (2025). Early Failure Detection for Air Production Unit in Metro Trains. In *Hybrid Artificial Intelligent Systems*, volume 14857, pages 339–351. Springer, Salamanca, Spain.
- [Zhang et al., 2022] Zhang, N., Graham, D. J., Bansal, P., and Hörcher, D. (2022). Detecting metro service disruptions via large-scale vehicle location data. *Transportation Research Part C: Emerging Technologies*, 144:103880.